[90]) rather than recall-based (like ROUGE [69]) as LLM-based methods tend to over-generate spans, which recall doesn't penalize. Our focus is measuring the overlap between generated and ground truth spans. Appendix A.2 provides a simplified example where General and Categorical Precision are computed, illustrating the suitability of these metrics for this evaluation setting.

	General			Categorical		
Expert Agreement	0.57			0.23		
Detector LLM	n=2	n=5	n=25	n=2	n=5	n=25
Claude3.5-Sonnet	0.43	**0.46**	0.44	0.20	0.21	0.20
Llama3.1-70b	0.42	0.45	0.38	0.16	0.21	0.14
GPT-4o	0.44	**0.46**	**0.46**	0.17	0.18	0.17

Table 9: Results of automated methods for detecting problematic spans in LLM-generated text, compared to agreement levels between three experts. Results report Precision scores for various LLMs used for detection, using a 2,5,25 examples in the few-shot prompt instruction, reporting both a General Precision and the stricter Categorical Precision.

We implement few-shot LLM-based methods [10] that have demonstrated competitive performance on tasks across several disciplines, often using fewer than 100 examples [10]. Our experiment varies the number of few-shot examples (2, 5, and 25) with the 2-shot prompt in Appendix A.1 and tests Llama3.1-70B, GPT-4o, and Claude3.5-Sonnet. As part of the collection of the LAMP Corpus, 50 paragraphs were edited independently, and we computed General and Categorical Precision on this set to estimate expert-expert agreement. Table 9 summarizes results. The best General Precision (0.46) is achieved by Claude-3.5 Sonnet and GPT-4o with a 5-shot prompt, below the expert agreement level (0.57). LLM-based methods can identify problematic spans with significant expert overlap, but improvement is possible. Performance improves from 2-shot to 5-shot prompts but plateaus thereafter (Claude-3.5 Sonnet and GPT-4o achieve similar or better performance with 5-shot vs. 25-shot prompts).

Categorical Precision is consistently lower than General Precision for both LLM-based methods and writer-writer comparison, suggesting that even when problematic spans are commonly identified, category agreement may differ. Section 5.2 explores this finding in detail, with examples of spans that could be reasonably assigned multiple categories. Table 10 illustrates the contrast between automatic editing and writer-selected edits by showing a paragraph with problematic spans and categories identified by a writer versus an LLM.

6.2 Automatic rewriting of problematic spans in any LLM-generated text

To propose improvements for detected problematic spans, we use few-shot prompting [10] with LLMs as well. We design prompts for each of the seven edit categories, incorporating examples of rewrites from writers. Each prompt includes a category definition, 25 examples from the LAMP Corpus with original paragraphs, a single problematic span for the category, the expert-proposed rewrite, and finally the input paragraph with the target span to be rewritten. The prompts for each category are listed in Appendix A.3.

The detection and rewriting methods can form a two-step pipeline for editing paragraphs. Detection identifies problematic spans and assigns categories while rewriting uses category-specific prompts to revise each detected span. A final step replaces all problematic spans in the original paragraph with their rewrites. Unlike the detection task, we do not evaluate the rewriting stage in isolation. Instead, we judge the complete pipeline that edits an entire paragraph (by detecting and rewriting multiple spans) through manual evaluation with 12 writers that annotated the LAMP Corpus. We describe this manual experiment next.

6.3 Evaluating Automatic Editing of LLM-generated Text

To evaluate editing quality, we design an evaluation task where participants read three variants of a paragraph and rank them in terms of overall preference: (1) an unedited **LLM-generated** paragraph from the LAMP Corpus, (2) the **Writer-edited** version from the LAMP Corpus, and (3) an **LLM-edited** version using our pipeline to detect and rewrite problematic spans. 6 out of 18 of our experts were unavailable during the preference annotations. Some of them were busy with their full-time jobs while others wanted to take a break after the edit task. Given the concern for proper turn around we re-hired 12 of the 18 experts who had participated in creating the LAMP Corpus for this evaluation.

We split the LLM-edited variant into two further sub-conditions:

- **Writer Detected and LLM Rewritten**: In this condition, the pipeline skips automatic detection of problematic spans, relying only on reference spans selected by the writer during manual editing. It runs solely the rewriting stage, simulating an oracle setting where problematic spans are manually provided. This condition is coded as **LLM-edited-Oracle**
- **LLM Detected and LLM Rewritten**: In this condition, the two-step pipeline is entirely automatic, with the automatically detected spans being provided to the automatic rewriting module. This condition fully automates editing of the paragraph and is coded as **LLM-edited-full**.

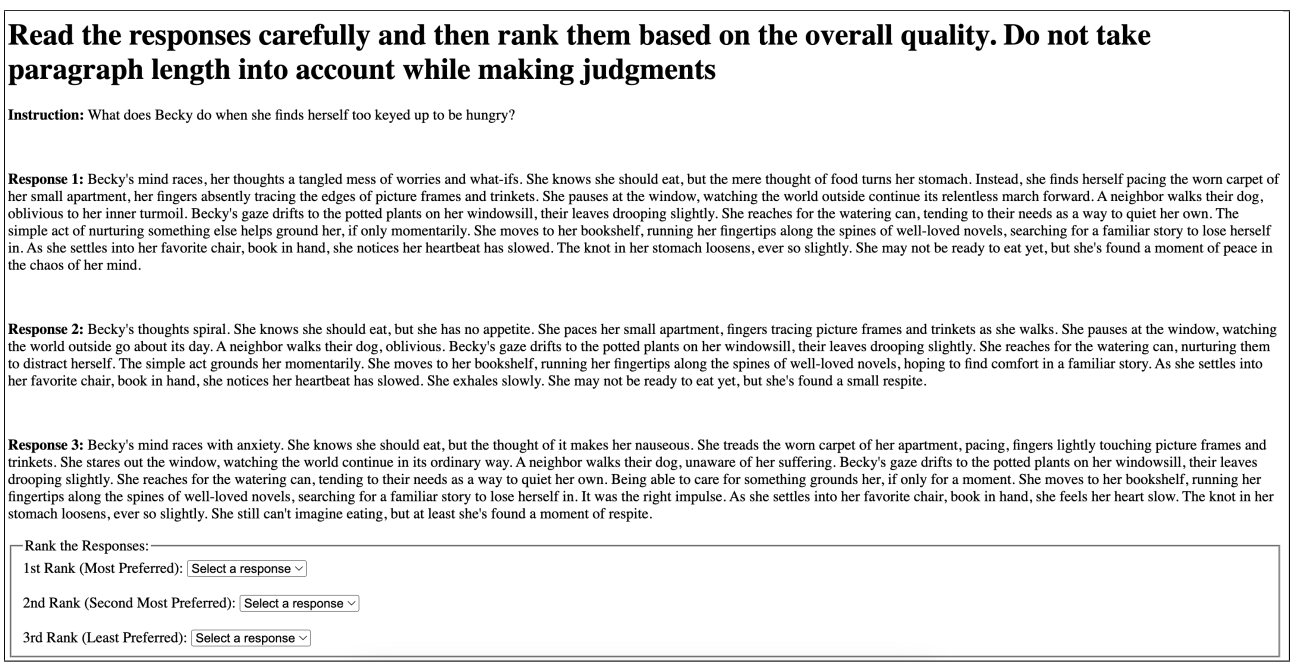
Table 11 shows examples of LLM-edited paragraphs under both sub-conditions. Although the examples here showcase 'Replacement' and 'Deletion' edits, analysis on a larger corpus of 200 LLM-edited paragraphs confirms that the automatic editing pipeline mirrors the edit type distribution of expert editors (shown in Figure 4a): roughly 65% 'replace' edits, 25% 'delete' edits, and 10% 'insert' edits. This surface-level analysis confirms the automatic pipeline's ability to mimic the editing process of experts measured in terms of relative proportions of character insertion and deletion.

In our pilot evaluation, we initially included all four conditions for annotation. However, ranking four paragraphs proved challenging for participants, especially when distinguishing between their second and third preferences. Based on this feedback, we redesigned the task to have participants judge only three conditions in each annotation. We always included the **LLM-generated** and **Writer-edited** paragraphs and alternated between including **LLM-edited-oracle** and **LLM-edited-full** paragraphs. While traditionally pairwise rankings tend to be easier for humans to annotate,

[10] Model fine-tuning experiments on LAMP Corpus are left for future work

Writer	Jackson leaned back in his office chair [, **staring out**$_{\text{Poor Sentence Structure}}$] through the expansive glass windows of the high-rise building. Below him, the city churned with [**relentless**$_{\text{Awkward Word Choice and Phrasing}}$] energy, [**a tapestry of modernity threaded with the hum of traffic,**$_{\text{Cliché}}$] the glow of digital billboards, [**and the unceasing flow of pedestrians. His desk, a**$_{\text{Poor Sentence Structure}}$] chaotic mixture of reports and coffee cups [**, contrasted sharply with the sleek orderliness of the cityscape**$_{\text{Poor Sentence Structure}}$] [**Despite the apparent advancement that framed his daily life, Jackson felt an inexplicable disconnection. His**$_{\text{Unnecessary/Redundant Exposition}}$] mind often wandered to an untouched box of old photographs at home, depicting simpler times – wooden houses, dirt paths, faces etched with stories of a bygone era. Those images spoke to him more than the polished chrome and glass now surrounding him.[**He couldn't help but yearn for the intangible warmth of community gatherings under open skies, unhurried conversations, and the tangible reality of a slower pace. Here, in the heart of progress, he was a mere observer, detached from the legacy beneath the steel and concrete, his own inclinations buried beneath the layers of modernity.**$_{\text{Cliché}}$]
LLM	Jackson leaned back in his office chair, staring out through the expansive glass windows of the high-rise building. Below him, the city [**churned with relentless energy**$_{\text{Cliche}}$].The city was [**a tapestry of modernity threaded with the hum of traffic, the glow of digital billboards, and the unceasing flow of pedestrians**$_{\text{Purple Prose}}$]. His desk, a [**chaotic mixture of reports and coffee cups**$_{\text{Cliche}}$] contrasted sharply with **the sleek orderliness.** [**of the cityscape**$_{\text{Cliche}}$]. Despite the apparent advancement that framed his daily life, Jackson felt **an inexplicable** [**disconnection**$_{\text{Cliche}}$] His mind often wandered to an untouched box of old photographs at home, depicting simpler times – [**wooden houses, dirt paths, faces etched with stories of a bygone era.**$_{\text{Cliche}}$] He couldn't help but yearn for [**the intangible warmth of community gatherings under open skies, unhurried conversations, and the tangible reality of a slower pace**$_{\text{Purple Prose)}}$] Here, in the [**heart of progress**$_{\text{Cliche}}$], he was a mere observer, [**detached from the legacy beneath the steel and concrete, his own inclinations buried beneath the layers of modernity.**$_{\text{Purple Prose)}}$]

Table 10: Example of problematic spans identified by a writer and by a LLM.



Read the responses carefully and then rank them based on the overall quality. Do not take paragraph length into account while making judgments

Instruction: What does Becky do when she finds herself too keyed up to be hungry?

Response 1: Becky's mind races, her thoughts a tangled mess of worries and what-ifs. She knows she should eat, but the mere thought of food turns her stomach. Instead, she finds herself pacing the worn carpet of her small apartment, her fingers absently tracing the edges of picture frames and trinkets. She pauses at the window, watching the world outside continue its relentless march forward. A neighbor walks their dog, oblivious to her inner turmoil. Becky's gaze drifts to the potted plants on her windowsill, their leaves drooping slightly. She reaches for the watering can, tending to their needs as a way to quiet her own. The simple act of nurturing something else helps ground her, if only momentarily. She moves to her bookshelf, running her fingertips along the spines of well-loved novels, searching for a familiar story to lose herself in. As she settles into her favorite chair, book in hand, she notices her heartbeat has slowed. The knot in her stomach loosens, ever so slightly. She may not be ready to eat yet, but she's found a moment of peace in the chaos of her mind.

Response 2: Becky's thoughts spiral. She knows she should eat, but she has no appetite. She paces her small apartment, fingers tracing picture frames and trinkets as she walks. She pauses at the window, watching the world outside go about its day. A neighbor walks their dog, oblivious. Becky's gaze drifts to the potted plants on her windowsill, their leaves drooping slightly. She reaches for the watering can, nurturing them to distract herself. The simple act grounds her momentarily. She moves to her bookshelf, running her fingertips along the spines of well-loved novels, hoping to find comfort in a familiar story. As she settles into her favorite chair, book in hand, she notices her heartbeat has slowed. She exhales slowly. She may not be ready to eat yet, but she's found a small respite.

Response 3: Becky's mind races with anxiety. She knows she should eat, but the thought of it makes her nauseous. She treads the worn carpet of her apartment, pacing, fingers lightly touching picture frames and trinkets. She stares out the window, watching the world continue in its ordinary way. A neighbor walks their dog, unaware of her suffering. Becky's gaze drifts to the potted plants on her windowsill, their leaves drooping slightly. She reaches for the watering can, tending to their needs as a way to quiet her own. Being able to care for something grounds her, if only for a moment. She moves to her bookshelf, running her fingertips along the spines of well-loved novels, searching for a familiar story to lose herself in. It was the right impulse. As she settles into her favorite chair, book in hand, she feels her heart slow. The knot in her stomach loosens, ever so slightly. She still can't imagine eating, but at least she's found a moment of respite.

Rank the Responses:

1st Rank (Most Preferred): Select a response

2nd Rank (Second Most Preferred): Select a response

3rd Rank (Least Preferred): Select a response

Figure 7: Interface used by participants to read through variants of a paragraph (one LLM-generated, one manually edited by an expert, one edited by an LLM-based system), and rank them in terms of preference.

three-way ranking allowed direct comparison between the key conditions we wanted to evaluate (LLM-generated v.s. Writer-edited v.s. LLM-edited), making the relative preferences clearer in a single

Writer Detected and LLM Rewritten	Dust settled on my window sill, ~~a quiet testament to the daily upheaval outside~~ **a reminder of the ongoing transformation outside**. Yellow machines trundled along the narrow street, ~~their relentless growl a constant soundtrack~~ **growling incessantly**. I watched the workers in their neon vests, ~~like bright insects against the concrete hive~~ **bright and busy against the concrete backdrop**, orchestrating a ballet of drills and hammers. Every so often, a stone ~~house crumbled~~ **mansion fell** to give way to sleek modernism, steel skeletons reaching shyly towards a haze-obscured sky. ~~The trees lining the avenue, long sentries guarding our history, stood fewer each day~~ **The trees lining the avenue dwindled each day**. Neighbors ~~nodded at one another with tight smiles~~ **exchanged tight smiles,** masking the void ~~of vanished gardens and familiar creeks~~ **left by vanished gardens**. ~~Change buzzed in the air, metallic and cold, seeping into conversations and dreams.~~ **Change felt cold and metallic, entering conversations and dreams**. I ~~found myself wondering if we were~~ **wondered if we were** building towards ~~something or away from it~~**something or abandoning it.**
LLM Detected and LLM Rewritten	~~Becky's mind races, her thoughts a tangled mess of worries and what-ifs.~~ **Becky's thoughts spiral**. She knows she should eat, but ~~the mere thought of food turns her stomach.~~**she has no appetite**. Instead, she ~~finds herself pacing the worn carpet of her small apartment, her fingers absently tracing the edges of picture frames and trinkets.~~ **paces her small apartment, fingers tracing picture frames and trinkets as she walks.** She pauses at the window, watching the world outside ~~continue its relentless march forward~~ **go about its day.** A neighbor walks their dog, ~~oblivious to her inner turmoil~~ **oblivious**. Becky's gaze drifts to the potted plants on her windowsill, their leaves drooping slightly. She reaches for the watering can, ~~tending to their needs as a way to quiet her own~~**nurturing them to distract herself.**~~The simple act of nurturing something else helps ground her, if only momentarily.~~ **The simple act grounds her momentarily.** She moves to her bookshelf, running her fingertips along the spines of well-loved novels, ~~searching for a familiar story to lose herself in~~ **hoping to find comfort in a familiar story.** As she settles into her favorite chair, book in hand, she notices her heartbeat has slowed. ~~The knot in her stomach loosens, ever so slightly.~~ **She exhales slowly.** She may not be ready to eat yet, but ~~she's found a moment of peace in the chaos of her mind.~~ **she's found a small respite.**

Table 11: Table showing LLM-edited paragraphs on both Oracle Writer predicted spans as well as LLM predicted spans

LLM-generated	Writer-edited	LLM-edited-full
2.55	1.47	1.99

LLM-generated	Writer-edited	LLM-edited-oracle
2.47	1.53	1.99

Table 12: Average Ranking across 600 preference judgments. LLM-edited > LLM-generated (p-value: 1.3e-11 for Writer Predicted spans; 2.8e-13 for LLM Predicted spans) and Writer-edited > LLM-generated (p-value: 1.1e-26 for Writer Predicted spans; 1.17e-31 for LLM Predicted spans) using Wilcoxon signed-rank test

evaluation rather than having to combine multiple pairwise comparisons. Additionally, we want to avoid any familiarity effects or biases that could arise from evaluators reading the same text multiple times, which would likely occur when collecting overlapping pairwise comparisons.

We note that to obtain automatic edits of a paragraph, we used the same LLM that had originally been used to generate the paragraph.[11] While not optimal, as a single LLM might offer slightly better detection and rewriting capabilities, this approach allows us to simplify the experiment conceptually and also test our hypothesis if edits lead to overall better alignment without relying on a single model family. We assess if using an LLM in a multi-stage pipeline (drafting, problem detection, rewriting) can enhance overall writing quality. Future work could potentially optimize this editing pipeline further, possibly yielding better results for LLM-edited conditions.

To ensure fairness, paragraph variants are displayed in a shuffled order and anonymized, and participants were not informed about the difference between the paragraphs (i.e. whether they are edited). For the curious reader, Figure 7 provides the interface used for the annotation task, including three variants of a paragraph. To conduct our experiments, we selected a total of 200 paragraph triplets (100 including an **LLM-edited-oracle** paragraph, and 100 including an **LLM-edited-full** paragraph) selecting samples from the LAMP Corpus's test set. Preference judgments were collected in batches of 25-35 paragraph triplets, with participants paid $35/hour. To account for potential subjectivity and calculate agreement and reliability, three experts judged each triplet, totaling 600 annotated

[11]In other words, we used GPT-4o in the two-step pipeline to generate edits to paragraphs that were originally generated by GPT-4o.